

Hrishabh Sahu

Hrishabhsahu123@gmail.com | +91 8103254557

[LinkedIn](#) | [GitHub](#) | [Portfolio Webpage](#)



Work Experience

Graduate Engineering Trainee (GET) at HCL Tech

Aug 2022 - Dec 2023

Worked with the SCCM tool in a US-based insurance company. Responsibilities included deploying MS Windows security patches, software updates to client VDIs, and troubleshooting errors. Received appreciation for outstanding contribution in JFM-23 Quarter.

1-Month Training at ETL Laboratory, Bhilai Steel Plant

Aug 2021 - Nov 2021

Observed the scheduled inspection and maintenance of numerical relays, PLCs, and circuit breakers. Learned about relay testing procedures, troubleshooting methods, and the role of PLCs in industrial automation.

Education

Year	Degree/Certificate	Institute	CPI / %
2025 - 2027	M.Tech / Telecomm Technology	Indian Institute of Technology, Delhi	8.375/10
2018 - 2022	B.Tech / Electrical & Electronics Engg.	Bhilai Institute of Technology, Durg	8.8/10
2018	Class XII	BSP Senior Secondary School Sector-7, Bhilai	80%
2016	Class X	BSP Senior Secondary School Sector-7, Bhilai	7.6/10

Projects

Implementation of error correcting code (LDPC) in 5G standard

Dec 2026 (ongoing)

Guide : Prof. Kaushik Saha | Indian Institute of Technology Delhi.

- Developed a full-cycle C++ software reference model for 5G NR LDPC channel coding compliant with the 3GPP TS 38.212 standard.
- Developed and verified the physical layer processing blocks, including the encoders, a Min-Sum decoder algorithm, rate matching, and rate recovery.
- Achieved a sequential baseline throughput of 0.25 Mbps to serve as the golden software reference for verification.
- Currently planning the Verilog RTL design phase to achieve high-speed, parallel hardware performance.

128Kb × 8 High Speed CMOS SRAM Design & Layout Implementation (Group Project)

Feb 2025 - Apr 2025

Guide : Prof. Kaushik Saha | Indian Institute of Technology Delhi.

- Developed the schematic and full-custom layout for a 3:8 block decoder, a 6:64 row decoder and an 8:256 column decoder with delays 59.67ps, 145.2ps and 174.9ps respectively.
- Sized and implemented the layout for the NMOS pass transistor logic and wordline drivers, successfully executing pitch-matching constraints to seamlessly interface with the memory core.
- Implemented a structural bitline twisting methodology after every 16 cells across the 512-wordline core array, to effectively mitigate bitline crosstalk.
- Utilized Euler's Path technique to optimize transistor routing topology, drastically minimizing diffusion breaks, flattening cell layout boundaries, and ensuring a 100% DRC/LVS clean peripheral block.
- Collaborated on the development of the asynchronous internal timing engine, incorporating a current-starved inverter loop (90ps delay) and a monostable multivibrator to generate a PVT-invariant 1.5ns master control pulse.
- Led the physical floorplanning, routing, and integration of the decoders, precharge blocks, and Charge Transfer Sense Amplifiers (CTSA).
- Involved in comprehensive post-layout parasitic extraction (PEX) simulations on the memory's worst-case critical path, validating reliable execution with a Read 0 delay of 5.36 ns and a Write 0 delay of 5.91 ns.
- Successfully achieved an optimized full-chip area layout of 2.574 mm² (inclusive of a 30% I/O pad overhead), yielding a core die area of 1.98 mm² and a 66% core efficiency that outperformed our estimated design constraints.

8-bit model CPU - RTL design and Timing Analysis (Vivado).

Oct 2025 - Dec 2025

Guide : Prof. Ankesh Jain | Indian Institute of Technology Delhi.

- Designed an 8-bit model CPU in Verilog with instruction fetch from 256-byte memory, arithmetic (add/sub), logical shifts, and unsigned 8-bit multiplication.
- Performed static timing analysis to determine maximum operating frequency.
- Simulated functional correctness with a comprehensive testbench and synthesized the design using Xilinx Vivado.
- Reduced critical-path latency by modular RTL design.

Wi-Fi-Based Home Automation (Self Project)

Jul 2023 - Aug 2023

- Used NodeMCU to remotely control electrical appliances.
- Used BLYNK app for controlling relay operation via a smartphone.

Demonstration of Vertical Axis Wind Turbine for Highways (Group Project)

Jan 2022 - Jun 2022

Guide : Asst. Prof. Abhijeet Lal | Bhilai Institute of Technology Durg.

- Developed a vertical axis wind turbine to harness wind turbulence from moving vehicles on highways.
- Connected the turbine to a DC motor with a 1:10 gear ratio and a battery charging circuit.
- Successfully demonstrated a renewable energy solution for power generation.

Skills

Languages : Verilog | C++ | Python (Beginner) | MATLAB | HTML.
Software : Cadence Virtuoso | Genus | Innovus | Calibre | Vivado | LT Spice | Microsoft Excel | Microsoft Word | SCCM | Service Now.
Courses Done : Semiconductor Memory Design | CMOS Digital IC Design | Digital Electronics | MIMO Wireless Communication | Digital Communication | Signal Processing Lab | Machine Learning | Signal Theory.
Ongoing Courses :

Achievements

- Scored **highest marks** in Semiconductor Memory Design course taken by Prof. Kaushik Saha among 30 students.
- Secured **97.06 percentile** in **GATE EE 2025** among candidates who appeared for the examination.
- Received appreciation for **outstanding contribution** during **JFM-23 Quarter** at **HCL Tech**.
- **Second position** in Academics in **Class XII**.